✪ Psychological Statistics ✪

Logistic Regression

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In [2]:
```
### [ Setup the working directory ]

setwd("/Users/wesley/[Course]/Python/R_Script")
getwd()
```

'/Users/wesley/[Course]/Python/R_Script'

In [3]:
```
### [ Loading the required libraries ]

library("dplyr")
library("rstatix")
library("ggpubr")
library("gmodels")
library("MASS")
library("mlogit")
```

---

(1) Logistic Regression

Example A: Bird & Isolation

[Hypothesis] The bird existence (0 or 1) could be predicted by the island size (area) and its isolation distance (isolation). *(2-tailed)*

- Null hypothesis H_0 **:** The bird existence has nothing to do with island area and isolation distance. $\rightarrow \beta = 0$
- Alternative hyp. H_1 **:** The bird existence can be predicted by island area and isolation distance. $\rightarrow \beta \neq 0$

In [19]:
```
### [ Step.1 ] Load data

island <- read.csv("isolation.csv")
attach(island)
head(island)
```

```
The following objects are masked from island (pos = 3):

    area, incidence, isolation


The following objects are masked from island (pos = 4):

    area, incidence, isolation
```

A data.frame: 6 × 3

	incidence	area	isolation
	<int>	<dbl>	<dbl>
1	1	7.928	3.317
2	0	1.925	7.554
3	1	2.045	5.883
4	0	4.781	5.932
5	0	1.536	5.308
6	1	7.369	4.934

In [2]:
```
### [ Step.2 ] Check assumptions

#----- (a) Outliers -----#
# influence.measures(model1) → check it after conducting logistic regression

#----- (b) Normality -----#
# Categorical Y does not fit normality assumption definitely.

#----- (c) Homogeneity of Variance -----#
# No variance estimation for categorical Y

# so we skip all those typical checks by now.
```

In [29]:
```
### [ Step.3-1 ] Logistic regression

#--> Modelling and
#model1 <- glm(incidence ~ 1, binomial)
#summary(model1)

model2 <- glm(incidence ~ area + isolation, binomial)
summary(model2)
```

```
Call:
glm(formula = incidence ~ area + isolation, family = binomial)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.8189  -0.3089   0.0490   0.3635   2.1192

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   6.6417     2.9218   2.273  0.02302 *
area          0.5807     0.2478   2.344  0.01909 *
isolation    -1.3719     0.4769  -2.877  0.00401 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 68.029  on 49  degrees of freedom
Residual deviance: 28.402  on 47  degrees of freedom
AIC: 34.402

Number of Fisher Scoring iterations: 6
```

In [28]:
```
### [ Step.3-2 ] compare different models

anova(model1,model2,test="Chi")
```

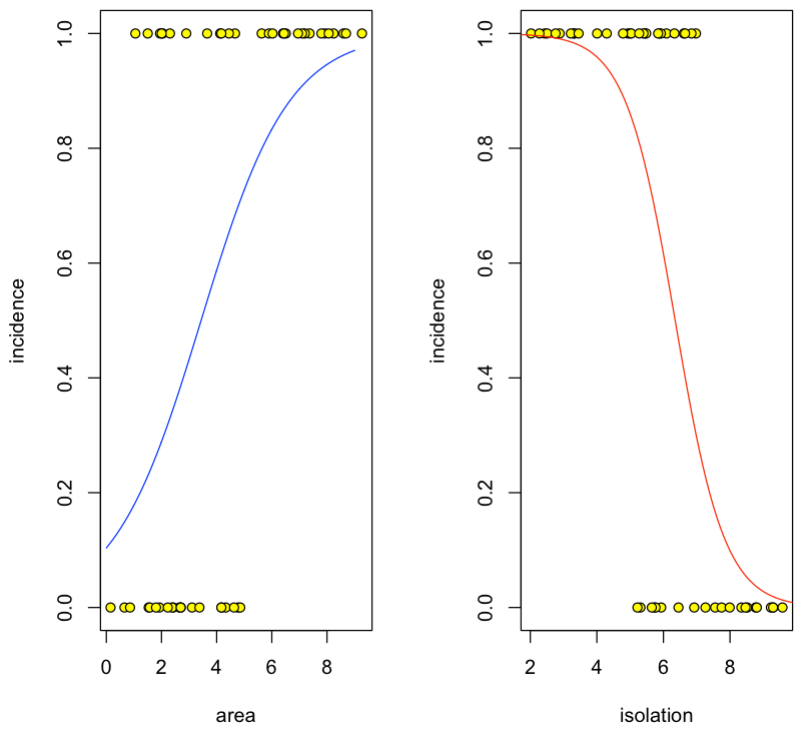
A anova: 2 × 5

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	49	68.02920	NA	NA	NA
2	47	28.40216	2	39.62704	2.483693e-09

In [44]:
```
### [ Step.4 ] Effect size: Pseudo R-square
# note: Odds ratio is only used for categorical X variable

round(pseudo.R2 <- (model2$null.deviance - model2$deviance)/model2$null.devianc
```

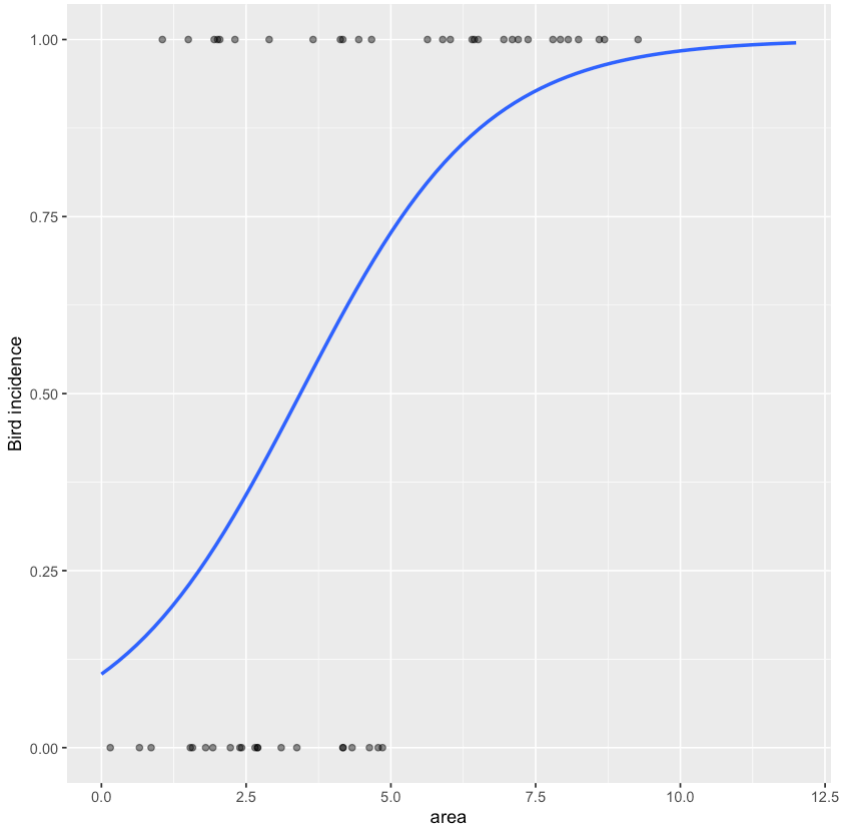
0.583

In [10]:
```
### [ Step.5 ] Visualization

#--> Plot relations btw incidence, area and isolation
modela <- glm(incidence~area,binomial)
modeli <- glm(incidence~isolation,binomial)

par(mfrow=c(1,2))
xv <- seq(0,9,0.01)
yv <- predict(modela,list(area=xv),type="response")
plot(area,incidence,pch=21,bg="yellow")
lines(xv,yv,col="blue")

xv2 <- seq(0,10,0.1)
yv2 <- predict(modeli,list(isolation=xv2),type="response")
plot(isolation,incidence,pch=21,bg="yellow")
lines(xv2,yv2,col="red")
```



In [74]:
```
ggplot(island, aes(x=area, y=as.numeric(incidence))) +
  geom_point(alpha=.5) +
  stat_smooth(method="glm", se=FALSE, fullrange=TRUE, method.args = list(family
  ylab("Bird incidence") + xlim(0,12)
```

```
`geom_smooth()` using formula 'y ~ x'
```



~ *Report* ~

- ***Both island size and isolation can predict the incidence of the special bird with Logistic regression, where $\chi^2_{(2)} = 39.63$, $p < .001$, pseudo $R^2 = 58.3\%$, AIC = 34.4.***
- The bird existence is positively associated with island size *($\beta = 0.58$, $p < .02$)*
- Instead, bird existence is negatively associated with isolation distance *($\beta = -1.37$, $p < .005$)*.

Example B: Pain Relief

[Hypothesis] Whether the pain can be cured (yes or no) was determined by the treatment and the duration of suffering. *(2-tailed)*

- Null hypothesis H_0 **:** The treatment does not provide the pain relief effect ($\beta = 0$).
- Alternative hyp. H_1 **:** The pain can be relived by the treatment ($\beta \neq 0$).

In [60]:
```
### [ Step.1 ] Data Loading

relief <-read.delim("Relief.dat", header=T)
#head(relief)

# put "Not Cured" as the baseline
levels(relief$Cured)
relief$Cured <- relevel(relief$Cured, "Not Cured")
levels(relief$Cured)

relief$Intervention <- relevel(relief$Intervention, "No Treatment")
levels(relief$Intervention)[2] <- "Treatment"
levels(relief$Intervention)

head(relief)
```

'Cured' · 'Not Cured'

'Not Cured' · 'Cured'

'No Treatment' · 'Treatment'

A data.frame: 6 × 3

	Cured	Intervention	Duration
	<fct>	<fct>	<int>
1	Not Cured	No Treatment	7
2	Not Cured	No Treatment	7
3	Not Cured	No Treatment	6
4	Cured	No Treatment	8
5	Cured	Treatment	7
6	Cured	No Treatment	6

```
           Intervention
Cured       No Treatment Treatment
  Not Cured           32        16
  Cured               24        41
```

In [75]:
```
### [ Step.2 ] Check assumptions

# we skip the typical checks by now.
```

In [39]:
```
### [ Step.3-1 ] Logistic regression

Model.1 <- glm(Cured ~ Intervention, data = relief, family = binomial())
summary(Model.1)

#Model.2 <- glm(Cured ~ Intervention + Duration, data = relief, family = binomi
#summary(Model.2)
```

```
Call:
glm(formula = Cured ~ Intervention, family = binomial(), data = relief)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.5940  -1.0579   0.8118   0.8118   1.3018

Coefficients:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)            -0.2877     0.2700  -1.065  0.28671
InterventionTreatment   1.2287     0.3998   3.074  0.00212 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 154.08  on 112  degrees of freedom
Residual deviance: 144.16  on 111  degrees of freedom
AIC: 148.16

Number of Fisher Scoring iterations: 4
```

In [40]:
```
### [ Step.3-2 ] The null deviance

Model.0 <- glm(Cured ~ 1, data = relief, family = binomial())
summary(Model.0)
```

```
Call:
glm(formula = Cured ~ 1, family = binomial(), data = relief)

Deviance Residuals:
   Min      1Q  Median      3Q     Max
-1.309  -1.309   1.052   1.052   1.052

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.3032     0.1903   1.593    0.111

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 154.08  on 112  degrees of freedom
Residual deviance: 154.08  on 112  degrees of freedom
AIC: 156.08

Number of Fisher Scoring iterations: 4
```

In [51]:
```
### [ Step.4-1 ] Evaluating residuals

# Interpreting residuals
# influence.measures(Model.1)

# Estimating pseudo R-square and p-value from log-likelihood
(LL.null <- Model.1$null.deviance)
(LL.model1 <- Model.1$deviance)
(pseudo.R2 <- (LL.null - LL.model1) / LL.null)
(pvalue <- 1-pchisq((LL.null - LL.model1), df=(length(Model.1$coefficients)-1))

# Alternative estimation (using "anova")
anova(Model.0, Model.1, test="Chi")
```

154.084004895399

144.157803678165

0.0644207114422539

0.00162942474293104

A anova: 2 × 5

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	112	154.0840	NA	NA	NA
2	111	144.1578	1	9.926201	0.001629425

In [54]:
```
### [ Step.4-2 ] Effect size - Odds Ratio

exp(Model.1$coefficients) %>% round(3)
exp(confint(Model.1)) %>% round(3)
```

(Intercept): 0.75 **InterventionTreatment:** 3.417

```
Waiting for profiling to be done...
```

A matrix: 2 × 2 of type dbl

	2.5 %	97.5 %
(Intercept)	0.437	1.269
InterventionTreatment	1.582	7.626

In [59]:
```
### [ Step.4-3 ] Grouping all statistics of interests

cbind(Beta=coef(Model.1), OR=exp(coef(Model.1)), CI=exp(confint(Model.1))) %>%
```

```
Waiting for profiling to be done...
```

A matrix: 2 × 4 of type dbl

	Beta	OR	2.5 %	97.5 %
(Intercept)	-0.28768	0.75000	0.43745	1.26867
InterventionTreatment	1.22867	3.41667	1.58201	7.62555

In [79]:
```
### [ Step.4-4 ] Pseudo R-squares (optional)

logisticPseudoR2s <- function(LogModel) {
    dev <- LogModel$deviance
    nullDev <- LogModel$null.deviance
    modelN <- length(LogModel$fitted.values)
    R.l  <- 1-dev/nullDev
    R.cs <- 1-exp(-(nullDev-dev)/modelN)
    R.n  <- R.cs/(1-(exp(-(nullDev/modelN))))
    cat("Pseudo R^2 for logistic regression\n")
    cat("==================================\n")
    cat("Hosmer and Lemeshow R^2 = ", round(R.l,3), "\n")
    cat("Cox and Snell R^2 = ", round(R.cs,3), "\n")
    cat("Nagelkerke R^2 = ", round(R.n,3), "\n")
}

logisticPseudoR2s(Model.1)
```

```
Pseudo R^2 for logistic regression
==================================
Hosmer and Lemeshow R^2 =  0.064
Cox and Snell R^2 =  0.084
Nagelkerke R^2 =  0.113
```

In [76]:
```
### [ Step.5 ] No Visualization, both X & Y are binary

xtabs(~Cured+Intervention, data = relief)
```

```
           Intervention
Cured       No Treatment Treatment
  Not Cured           32        16
  Cured               24        41
```

~ *Report* ~

- ***The intervention can predict the pain relief ($\chi^2_{(1)}$ = 9.93, $p < .01$, pseudo R^2 = 6%, AIC = 148.16).***
- With the treatment, the odds of being cured will be enhanced by 3.4 times higher (341%) *($p < .01$)*

---

(2) Testing Assumptions of Logistic Regression

Example: Penalty Kick

[Hypothesis] **The success of penalty kick (missed or scored) could be affected by the anxiety, worriness and the success rate of past penalty kick.** ***(2-tailed)***

- Null hypothesis ***H_0 :*** No prominent relations can be observed among success of penalty kick and the 3 factors (all βs = 0).
- Alternative hyp. ***H_1 :*** The success of penalty kick can be predicted by at least one of the 3 factors (βs ≠ 0).

In [82]:
```
### [ Step.1 ] Data Loading

penaltyData<-read.delim("penalty.dat", header = TRUE)
head(penaltyData,5)
```

A data.frame: 5 × 4

	PSWQ	Anxious	Previous	Scored
	<int>	<int>	<int>	<fct>
1	18	21	56	Scored Penalty
2	17	32	35	Scored Penalty
3	16	34	35	Scored Penalty
4	14	40	15	Scored Penalty
5	5	24	47	Scored Penalty

In [97]:
```
### [ Step.3 ] Logistic regression

penalty.0 <- glm(Scored ~ 1, data = penaltyData, family = binomial())
summary(penalty.0)

penalty.1 <- glm(Scored ~ Previous + PSWQ, data = penaltyData, family = binomia
summary(penalty.1)

penalty.2 <- glm(Scored ~ Previous + PSWQ + Anxious, data = penaltyData, family
summary(penalty.2)

#This works too:
# penalty.2 <- update(penalty.1, .~. + Anxious)
```

```
Call:
glm(formula = Scored ~ 1, family = binomial(), data = penaltyData)

Deviance Residuals:
   Min      1Q  Median      3Q     Max
-1.235  -1.235   1.121   1.121   1.121

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.1335     0.2315   0.577    0.564

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 103.64  on 74  degrees of freedom
Residual deviance: 103.64  on 74  degrees of freedom
AIC: 105.64

Number of Fisher Scoring iterations: 3


Call:
glm(formula = Scored ~ Previous + PSWQ, family = binomial(),
    data = penaltyData)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.2212  -0.3306   0.1038   0.5046   1.6067

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.28031    1.67017   0.767  0.44333
Previous     0.06480    0.02209   2.934  0.00335 **
PSWQ        -0.23009    0.07983  -2.882  0.00395 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 103.638  on 74  degrees of freedom
Residual deviance:  48.662  on 72  degrees of freedom
AIC: 54.662

Number of Fisher Scoring iterations: 6
```

```
Call:
glm(formula = Scored ~ Previous + PSWQ + Anxious, family = binomial(),
    data = penaltyData)

Deviance Residuals:
     Min        1Q    Median        3Q       Max
-2.31374  -0.35996   0.08334   0.53860   1.61380

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -11.49256   11.80175  -0.974  0.33016
Previous      0.20261    0.12932   1.567  0.11719
PSWQ         -0.25137    0.08401  -2.992  0.00277 **
Anxious       0.27585    0.25259   1.092  0.27480
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 103.638  on 74  degrees of freedom
Residual deviance:  47.416  on 71  degrees of freedom
AIC: 55.416

Number of Fisher Scoring iterations: 6
```

In [101]:

```
### [ Step.4 ] Effect size

# Compute odds ratio
exp(penalty.1$coefficients)
exp(confint(penalty.1))

# compare model1 and model 2
anova(penalty.1, penalty.2, test="Chi")
```

(Intercept): 3.59775997097828 **Previous:** 1.06694818062729 **PSWQ:** 0.794458377100457

```
Waiting for profiling to be done...
```

A matrix: 3 × 2 of type dbl

	2.5 %	97.5 %
(Intercept)	0.1379603	108.7408589
Previous	1.0263545	1.1216277
PSWQ	0.6614590	0.9115488

A anova: 2 × 5

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	72	48.66180	NA	NA	NA
2	71	47.41554	1	1.246267	0.2642667

In [86]:
```
### [ Step.2-1 ] Check assumptions - Multicollinearity -

car::vif(penalty.1)
car::vif(penalty.2)

cor(penaltyData[, c("Previous", "PSWQ", "Anxious")])

# so it's better removing one predictor with high correlation
```

Previous: 1.01237203645737 **PSWQ:** 1.01237203645737

Previous: 35.2271130378127 **PSWQ:** 1.08976661632689 **Anxious:** 35.5819755134594

A matrix: 3 × 3 of type dbl

	Previous	PSWQ	Anxious
Previous	1.0000000	-0.6435448	-0.9928699
PSWQ	-0.6435448	1.0000000	0.6516416
Anxious	-0.9928699	0.6516416	1.0000000

In [88]:
```
### [ Step.2-2 ] Check assumptions - the linearity of the logit -

#Create the interaction of PSWQ with log(PSWQ)
penaltyData$logPSWQInt<-log(penaltyData$PSWQ)*penaltyData$PSWQ

#Create the interaction of Anxious and Previous with their logs

penaltyData$logAnxInt<-log(penaltyData$Anxious)*penaltyData$Anxious
penaltyData$logPrevInt<-log(penaltyData$Previous)*penaltyData$Previous

head(penaltyData)

penaltyTest.1 <- glm(Scored ~ PSWQ +
                       Anxious +
                       Previous +
                       logPSWQInt +
                       logAnxInt +
                       logPrevInt,
                     data=penaltyData, family=binomial())
summary(penaltyTest.1)
```

A data.frame: 6 × 7

	PSWQ	Anxious	Previous	Scored	logPSWQInt	logAnxInt	logPrevInt
	<int>	<int>	<int>	<fct>	<dbl>	<dbl>	<dbl>
1	18	21	56	Scored Penalty	52.02669	63.93497	225.41969
2	17	32	35	Scored Penalty	48.16463	110.90355	124.43718
3	16	34	35	Scored Penalty	44.36142	119.89626	124.43718
4	14	40	15	Scored Penalty	36.94680	147.55518	40.62075
5	5	24	47	Scored Penalty	8.04719	76.27329	180.95694
6	1	15	67	Scored Penalty	0.00000	40.62075	281.71441

```
Call:
glm(formula = Scored ~ PSWQ + Anxious + Previous + logPSWQInt +
    logAnxInt + logPrevInt, family = binomial(), data = penaltyData)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.0685  -0.3846   0.1116   0.5460   1.8272

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.87885   14.92410  -0.260    0.795
PSWQ        -0.42233    1.10267  -0.383    0.702
Anxious     -2.64485    2.79702  -0.946    0.344
Previous     1.66601    1.48202   1.124    0.261
logPSWQInt   0.04393    0.29675   0.148    0.882
logAnxInt    0.68077    0.65277   1.043    0.297
logPrevInt  -0.31855    0.31731  -1.004    0.315

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 97.283  on 70  degrees of freedom
Residual deviance: 45.909  on 64  degrees of freedom
  (4 observations deleted due to missingness)
AIC: 59.909

Number of Fisher Scoring iterations: 7
```

In [105]:
```
### [ Step.5-1 ] Confirm which model to be used (Scored ~ PSWQ + Previous)

anova(penalty.0, penalty.1, test="Chi")
logisticPseudoR2s(penalty.1)
cbind(Beta=coef(penalty.1), OR=exp(coef(penalty.1)), CI=exp(confint(penalty.1))
```

A anova: 2 × 5

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	74	103.6385	NA	NA	NA
2	72	48.6618	2	54.97669	1.153355e-12

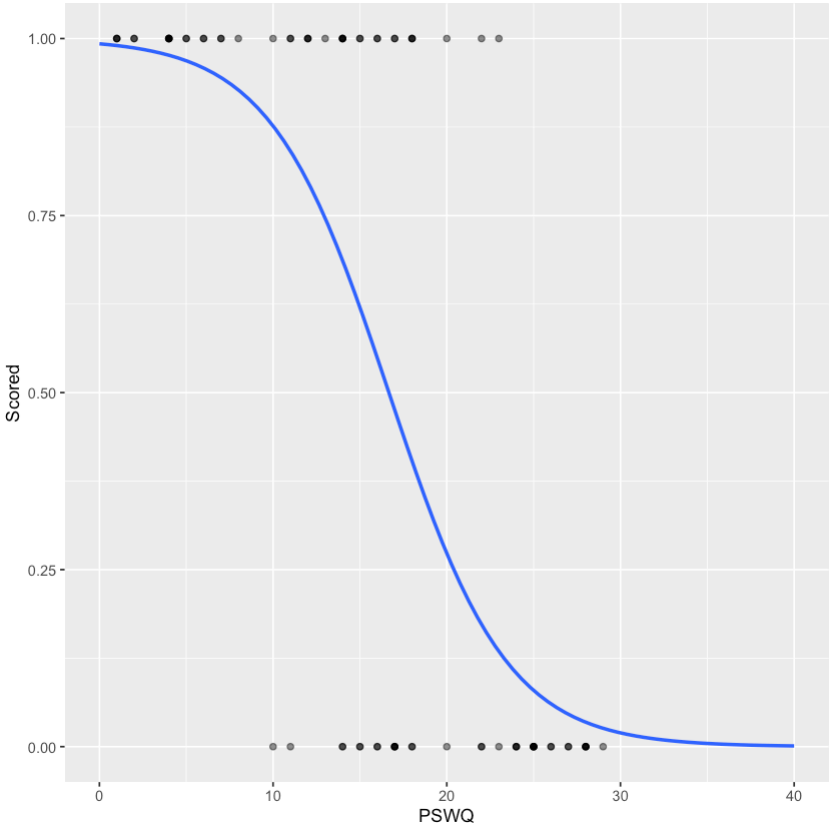
```
Pseudo R^2 for logistic regression
==================================
Hosmer and Lemeshow R^2 =  0.53
Cox and Snell R^2 =  0.52
Nagelkerke R^2 =  0.694

Waiting for profiling to be done...
```

A matrix: 3 × 4 of type dbl

	Beta	OR	2.5 %	97.5 %
(Intercept)	1.28031	3.59776	0.13796	108.74086
Previous	0.06480	1.06695	1.02635	1.12163
PSWQ	-0.23009	0.79446	0.66146	0.91155

In [103]:
```
### [ Step.5-2 ] Visualization (Scored ~ PSWQ + Previous)

ggplot(penaltyData, aes(x=PSWQ, y=as.numeric(Scored) - 1)) +
  geom_point(alpha=.5) +
  stat_smooth(method="glm", se=FALSE, fullrange=TRUE,
              method.args = list(family=binomial)) +
  ylab("Scored") + xlim(0, 40)
```

```
`geom_smooth()` using formula 'y ~ x'
```



~ *Report* ~

- ***The penalty-kick success rate can be predicted by the worriness and the past history of penalty kick ($\chi^2_{(2)}$ = 54.98, p < .001, pseudo R^2 = 53%, AIC = 54.66).***
- With one point increment in past success rate, the penalty-kick success rate will increase by 6.7% *(p < .005)*. [because OR = 1.06695]
- With one point increment in PSWQ, the penalty-kick success rate will decrease by 21% *(p < .005)*. [because OR = 0.79446]